Electronic Signature Information

Name	DOC2659998
Revision	1
Type	Controlled Document
Title	MR2022H2 Host Software SSDD
Originator	305019292_balachandar__g k
Release Date	09/07/2022 07:58:44 PM
Obsolete Date	

File Name	File Description	File Size (Bytes)
MR2022H2 Host Software SSDD.docx	update doc	21030724

Route	Signer	Function	Status	Comments	Completion Date
R-10529846	305002209_shivabasappa_r_channashetti		Approve	Approved as a Task Assignee	7 Sep 2022 19:57:14 GMT

Periodic Review

There are no signatures or routes related to this business object.

Obsolesence Approval

There are no signatures or routes related to this business object.

* Printed versions are For Reference Only *
+ Indicates a task was reassigned from an original assignee

© 2022 by General Electric Company. All Rights Reserved.

Table of Contents

1. Introduction 7

1.1 Purpose and Scope 7

1.2 Document Overview 7

1.3 Applicable Documents 8

1.3.1 Referenced Documents 8

1.3.2 Supporting Information 9

2. High Level Architectural Design 10

2.1 Design Overview 11

2.2 Functional Description 12

2.2.1 Rx UI 12

2.2.2 Graphic Rx UI 12

2.2.3 Geometry 12

2.2.4 Acquisition UI 12

2.2.5 State Machine 12

2.2.6 Image Receive Chain 12

2.2.7 External Interface 13

2.2.8 Coil Data 13

2.2.9 Gating UI 13

2.3 External Software Interfaces 13

2.4 Hardware Interfaces 15

2.5 User Interfaces 15

2.6 Architectural Error Handling and Recovery 17

2.7 Architectural Risk Control 18

2.7.1 Software Segregation: RF Power Protection 19

2.7.2 Software Segregation: Image Flip Protection 21

2.7.3 Software Segregation: Patient Identification Protection 22

2.8 Existing Design Reusability 23

2.9 Future Considerations 24

* Printed versions are For Reference Only *
Before using this document, consult MyWorkshop for the latest revision. GE Proprietary and Confidential Information.
Template Reference: 7.3.3-03MRF01_Design Description Template_DOC0600591

© 2022 by General Electric Company. All Rights Reserved.

3. Design Architecture.....24

3.1 Internal Functional or Object design overview.....25

3.1.1 Lx Scan.....31

3.1.2 Mx JVM.....31

3.1.3 CIM.....**Error! Bookmark not defined.**

3.1.4 LAIS/ATP/SVAT/ILT.....31

3.1.5 Prescan & SpectroPS.....32

3.1.6 GRX Server (2 sets).....32

3.1.7 GRx Proxy.....32

3.1.8 Autoview.....32

3.1.9 DS Server.....32

3.1.10 BW Proxy.....33

3.1.11 RXM.....33

3.1.12 BWRT.....33

3.1.13 PXE.....33

3.1.14 IFCC.....33

3.1.15 TIR.....33

3.1.16 IST.....34

3.1.17 APP.....34

3.1.18 Auto Post Processing.....34

3.1.19 RTIA.....34

3.1.20 Display Server.....34

3.1.21 RIR.....35

3.1.22 RTPSI.....35

3.1.23 NAV.....35

3.1.24 HGP, WaveRec & WaveDisp.....35

3.1.25 In Room Display.....35

3.1.26 Auto Voice.....35

3.1.27 Power Monitor.....36

3.1.28 High Order Shim.....36

© 2022 by General Electric Company. All Rights Reserved.

3.1.29 TPS/IPG Emulator (TIE) 36
3.1.30 Localizer Viewer 36
3.1.31 Express Spine Plugin 36
3.1.32 One Touch Spine Library 37
3.1.33 External Scanner Interface (EXSI) 37
3.1.34 Dual Drive 37
3.2 Risk Assignments 37
3.3 Internal Software Interfaces 37
3.3.1 Lx Scan Communications 40
3.3.2 Mx JVM Communications 42
3.3.3 CIM Communications 43
3.3.4 SVAT Communications 43
3.3.5 Prescan & SpectroPS Communications 43
3.3.6 GRX Server (set 1) Communications 44
3.3.7 GRX Server (set 2) Communications 44
3.3.8 GRx Proxy Communications 45
3.3.9 Autoview Communications 45
3.3.10 DS Server Communications 45
3.3.11 BW Proxy Communications 46
3.3.12 IFCC Communications 46
3.3.13 TIR Communications 46
3.3.14 IST Communications 47
3.3.15 APP Communications 47
3.3.16 RTIA Communications 47
3.3.17 Display Server Communications 48
3.3.18 RIR Communications 48
3.3.19 RTPSI Communications 48
3.3.20 ILT Communications 49
3.3.21 LAIS Communications 49
3.3.22 NAV Communications 49

* Printed versions are For Reference Only *
Before using this document, consult MyWorkshop for the latest revision. GE Proprietary and Confidential Information.

© 2022 by General Electric Company. All Rights Reserved.

3.3.23 HGP, WaveRec & WaveDisp Communications 49
3.3.24 AutoVoice Communications 50
3.3.25 In Room Monitor 50
3.3.26 EXSI 50
3.4 Processing 51
3.4.1 Key Processing 51
3.4.2 Preprocessing 58
3.4.3 Initialization 59
3.4.4 Termination 59
3.4.5 Error Handling and Recovery 59
3.5 Data Design 59
3.5.1 Static Data 59
3.5.2 Dynamic Data 60
4. Detailed Design 60
4.1 Lx Scan 60
4.2 Mx JVM 60
4.3 CIM 61
4.4 SVAT 61
4.5 Prescan & SpectroPS 61
4.6 GRX Server (2 sets) and Localizer Viewer 61
4.7 GRx Proxy 61
4.8 Autoview 61
4.9 DS Server 61
4.10 BW Proxy 61
4.11 RXM 61
4.12 BWRT 61
4.13 IFCC 62
4.14 TIR 62
4.15 IST 62
4.16 APP 62

© 2022 by General Electric Company. All Rights Reserved.

4.17 RTIA 62
4.18 Display Server 62
4.19 RIR 62
4.20 RTPSI 62
4.21 ILT 62
4.22 NAV 62
4.23 Auto Voice 62
4.24 Power Monitor 63
4.25 In Room Monitor 63
4.26 Express Spine Plugin 63
4.27 One Touch Spine Library 63
4.28 EXSI 63
4.29 Dual Drive 63
5. Software Generation 63
5.1 Tools Used 63
5.1.1 UIMX UI Builder 63
5.1.2 TAO IDL Compiler 63
5.1.3 OPENORB IDL Compiler 64
5.1.4 JAVA Compiler 64
5.2 Files Described 64
5.3 SOUP Used 67
5.3.1 JAVA 67
5.3.2 ACE/TAO, Open ORB 67
5.3.3 JGOODIES 68
5.3.4 JGo 68
5.3.5 XERCES 68
5.3.6 XRT 68
5.3.7 Merge DICOM ToolKit 68
6. Traceability 68
7. Glossary 68

© 2022 by General Electric Company. All Rights Reserved.

1. Introduction

This document describes the system level design of the Host Software subsystem. The Host Software comprises the prescription, image generation, and host side controls for the MR scanners acquisition software.

1.1 Purpose and Scope

This document describes the interfaces and processing associated with the Host Software Subsystem. The document is intended to provide software developers with the design details necessary to understand the overall design of the Host Software Subsystem. This document provides the next level of detail after the MR Software Architecture Design Document, Reference [1]. It must be used in concert with the other SSDD level documents to fully understand the overall design of the MR Software System. Reference [2] provides the subsystem level requirements for both the Host Software subsystem and the Host Software Platform subsystem. The software runs under the operating system defined for the host hardware, which is covered in the SWADD, Reference [1].

The MR Host software is common across multiple products based on a wide range of hardware configurations. The hardware specific components are enabled via system configuration or install selections or by the presence of the hardware components. Details of these configuration details are contained in the detailed design for the associated components. Not all features will be active on all systems although the basic software support is present. Refer to the specific product configurations for product specific hardware configuration details.

1.2 Document Overview

This software design document is intended to provide a high level design view of the MR Host Software subsystem:

- With sufficient detail to expose all major host SW units.
- Including a high level overview of inter-unit operations.
- Including all significant operating modes and states.
- Reference lower level design documents for software unit details.

Details are at a high enough level to allow the units to evolve internally without impacting this document.

With this in mind, Section 2 provides a high level design of the host software showing the relationship between the hardware, software and components external to this subsystem. Section 3 provides the functional overview of the individual software units, the interactions between the units and other software subsystems and major processing flows within the MRI host subsystem. Section 4 provides the references to the unit level documents providing the detailed design information.

Released

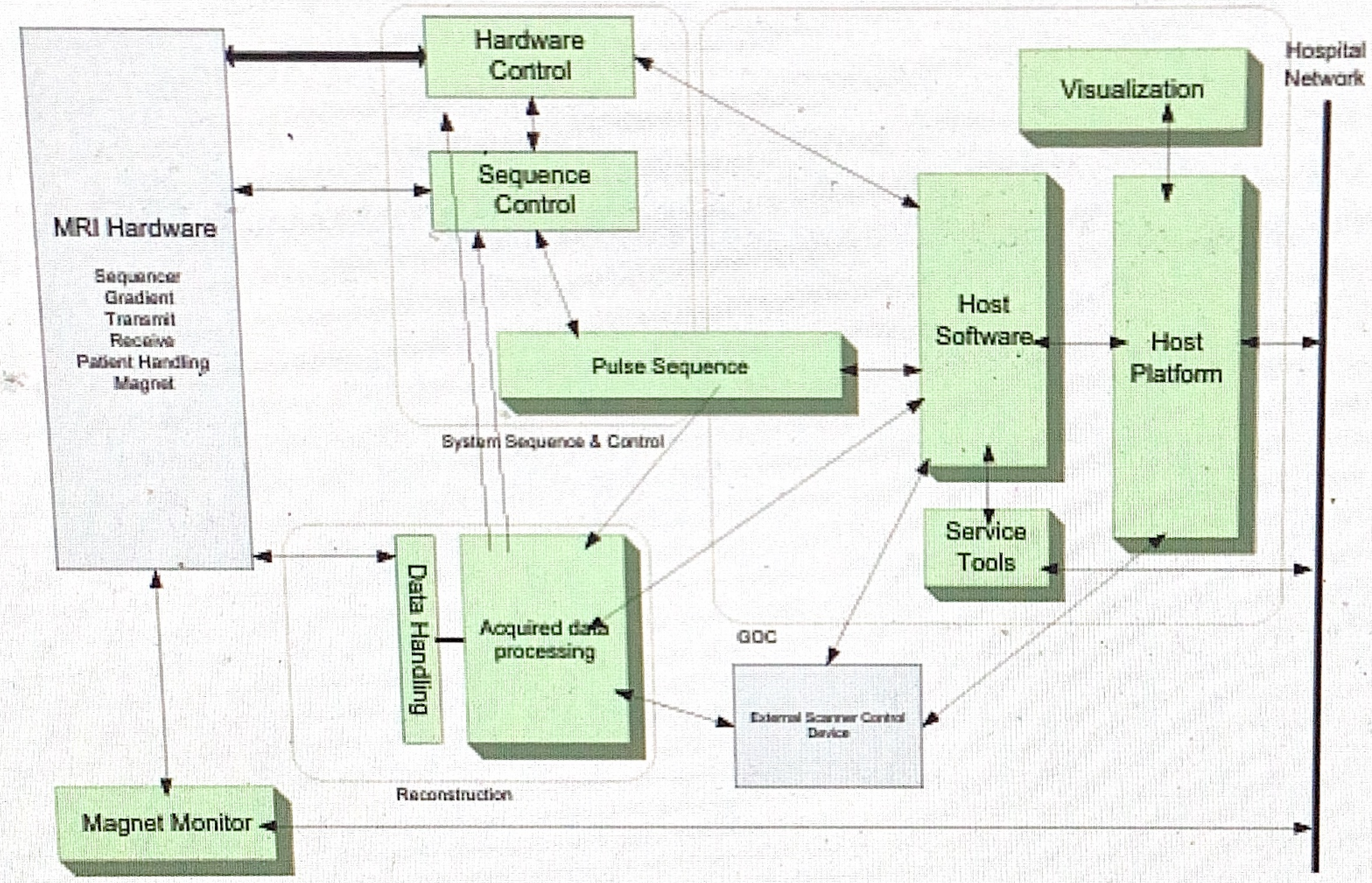
* Printed versions are For Reference Only *
Before using this document, consult MyWorkshop for the latest revision. GE Proprietary and Confidential Information.
Template Reference 7.3.3-03MREF01 Design Description Template DOC2659591

© 2022 by General Electric Company. All Rights Reserved.

2. High Level Architectural Design

The encompassing upper level design for this document is named: Rio Software Architecture Design Document (SWADD), Reference [1]. Section 3.1.1 provides the high level architectural overview for the Host Software subsystem. Figure 1 shows how the host software fits within the overall software architecture.

Figure 1 Software Subsystems



The host software component communicates with the following other subsystems: Host Platform, Service Tools, Acquired Data Processing, Pulse Sequence, Hardware Control, and External Scanner Control Device.

Released

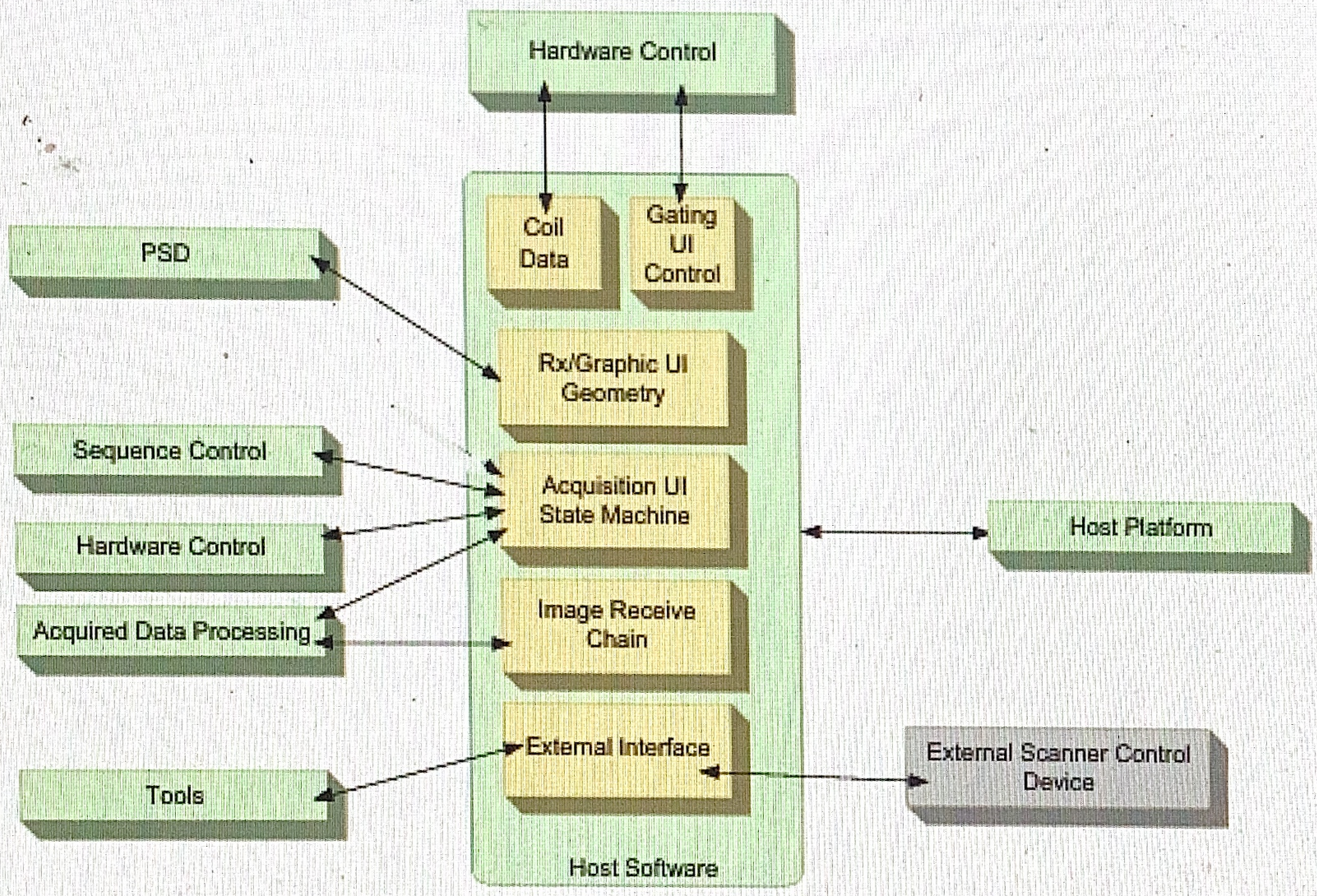
* Printed versions are For Reference Only *
Before using this document, consult MyWorkshop for the latest revision. GE Proprietary and Confidential Information.
Template Reference: 7.3.3-03MRF01_Design Description Template_DOC0600591

© 2022 by General Electric Company. All Rights Reserved.

2.1 *Design Overview*

The logical layout for the host software is shown in the Figure 2 below. The software is divided into major components shown within the Host Software block. This figure also shows the primary communication connections with other subsystems within the MR Software System. The user will interact with the software at various levels through the user interface; however, not all areas are exposed directly to the user.

Figure 2 Host Software Subsystem in Relationship with other Software Subsystems



* Printed versions are For Reference Only *
Before using this document, consult MyWorkshop for the latest revision. GE Proprietary and Confidential Information.
Template Reference: 7.3.3-03MRF01_Design Description Template_DOC0600591

Released

© 2022 by General Electric Company. All Rights Reserved.

2.2 *Functional Description*

The paragraphs below further describe each of the host software components depicted in Figure 2.

2.2.1 Rx UI

The RX UI component is responsible for displaying & controlling prescription UI. Based on contents of protocol, it launches appropriate PSD. It communicates parameter values and hardware configuration details to PSD. It queries the information from PSD and uses it to build user interface. When operator modifies parameter, it communicates with PSD to determine parameter validity. It also provides user interface for selecting coil from the list of connected and installed coils. It also provides application launch UI that lists all pulse sequences and imaging options.

2.2.2 Graphic Rx UI

The Graphic Rx UI component enables graphical way to prescribe scanning range parameters. After exam is started, once localizers are scanned it loads images into viewport. It allows user to prescribe ROI for acquisition, sat bands, shimming, tracking etc. It also provides common viewport capabilities like window width window level adjustment, zoom, report cursor etc. It also displays coil extents of selected coil.

2.2.3 Geometry

The geometry components Is responsible for converting data between coordinate systems. The basic three coordinate systems are mechanical, table and patient. Patient coordinate system is used by operator to input the data and for annotation on image. The mechanical and table coordinate systems are used internally to control the table positioning and spatial location of the acquisition volumes.

2.2.4 Acquisition UI

The Acquisition UI component is responsible for displaying & controlling acquisition UI. This is used by real time, prescan and spectro application. This communicates user input to AGP & VRE and displays response.

2.2.5 State Machine

The State Machine component is responsible for managing state of the system. System goes through different state from prescription to scan done. This component controls scanner UI based on the state. It also enables automatic/manual change of state based on user choice.

2.2.6 Image Receive Chain

The image receive chain component is responsible for generating, displaying & installing images. Based on user prescription it initially generates image header. Then it combines images header with pixel data from recon. Finally it generates DICOM image using translator. This image is then displayed automatically in image viewer and is submitted for installation

* Printed versions are For Reference Only *
Before using this document, consult MyWorkshop for the latest revision. GE Proprietary and Confidential Information
Template Reference: 7.3.3-03MRF01_Design Description Template; DOC0600591

© 2022 by General Electric Company. All Rights Reserved.

2.2.7 External Interface

The external interface component provides programmatic interface to the scanner. Remote control capability of this is used by focused ultrasound. Scripting capability is used by system tools in calibrating the system.

2.2.8 Coil Data

The coil data management facility comprises the software that controls access to the coil database, process coil codes, provides coil data to coil data consumers as well as compute the settings necessary to operate the system using connected coils. For the GEM coil set, the coil data management manages the data necessary to locate floating coils and to generate the data necessary to drive the automatic coil selection and automatic calibration series generation. For dynamic coil configuration, it provides new clinical coil based on selected Receive Element Groups (REG) and element data by the AutoCoil Selection Algorithm.

2.2.9 Gating UI

The Gating UI component provides the interface and control functions to setup the cardiac and respiratory gating sub-system as well as the waveform display.

2.3 External Software Interfaces

The host software subsystem connects to the following subsystems that are not part of the host software subsystem:

- Host platform
- Hardware Control
- Sequence Control
- Pulse Sequence (PSD)
- Acquired Data Processing
- External Scanner Control Device
- Tools

Except for communications to the external scanner control device, all communications are considered within the MRI system and do not have any security concerns. The communication channel is to the external scanner control device requires authentication prior to exchanging MRI commands or data.

The Table 1 below shows the major communications between the Host Software and the listed subsystems.

Table 1 External Software Communications

© 2022 by General Electric Company. All Rights Reserved.

From	To	Data	Interface Standard
Host Software	Pulse Sequence	user selection	message-Q
Pulse Sequence	Host Software	UI/Image/recon information	message-Q
Host Software	Hardware Control	User selected gating options	MrMail
Hardware Control	Host Software	Gating Waveform Data	MrMail
Hardware Control	Host Software	Landmark Position	MrMail
Hardware Control	Host Software	Connect/Disconnect Coil	MrMail
Host Software	Hardware Control	Table Position	MrMail
Host Software	Hardware Control	Hardware Settings	MrMail
Hardware Control	Host Software	Scanner State Start/Stop/Pause	MrMail
Host Software	Hardware Control	Boot-up Packet	TCP/IP
Acq. Data Proc.	Host Software	prescan frames	TCP/IP & MrMail
Acq. Data Proc.	Host Software	image data	MrMail & file server
Host Software	Acq. Data Proc.	prescription data	TCP/IP & MrMail
Host Platform	Host Software	Start Exam, Setup, Coil Selection, Scan operations, Copy/Paste	GEOCOM
Host Software	Host Platform	Scan status, Recon status	GEOCOM
Host Software	Host Platform	DICOM images	File I/O and messages
Host Software	Host Platform	system messages	TCP/IP
Host Platform	Host Software	DICOM images	File I/O and messages
Service Tools	Host Software	SVAT Command + data	SVAT
Host Software	Service Tools	Command Status + requested data	SVAT
Host Software	Ext. Scanner Ctrl.	image data	TCP/IP
Host Software	Ext. Scanner Ctrl.	responses to requests (status & data)	TCP/IP
Ext. Scanner Ctrl.	Host Software	commands and status requests	TCP/IP
Ext. Scanner Ctrl.	Host Software	image data registration	TCP/IP
Host Software	PXE	Query	TCP/IP
PXE	Host Software	Hardware Paradigms	TCP/IP
Host Software	Sequence Control	pulse sequence download request	MrMail

Although all of the processes run concurrently, synchronization is managed via message exchange (send/response) or event notification.

* Printed versions are For Reference Only *
Before using this document, consult MyWorkshop for the latest revision. GE Proprietary and Confidential Information.
Template Reference: 7.3.3-03MRF01_Design Description Template_DOC0600591

© 2022 by General Electric Company. All Rights Reserved.

2.4 Hardware Interfaces

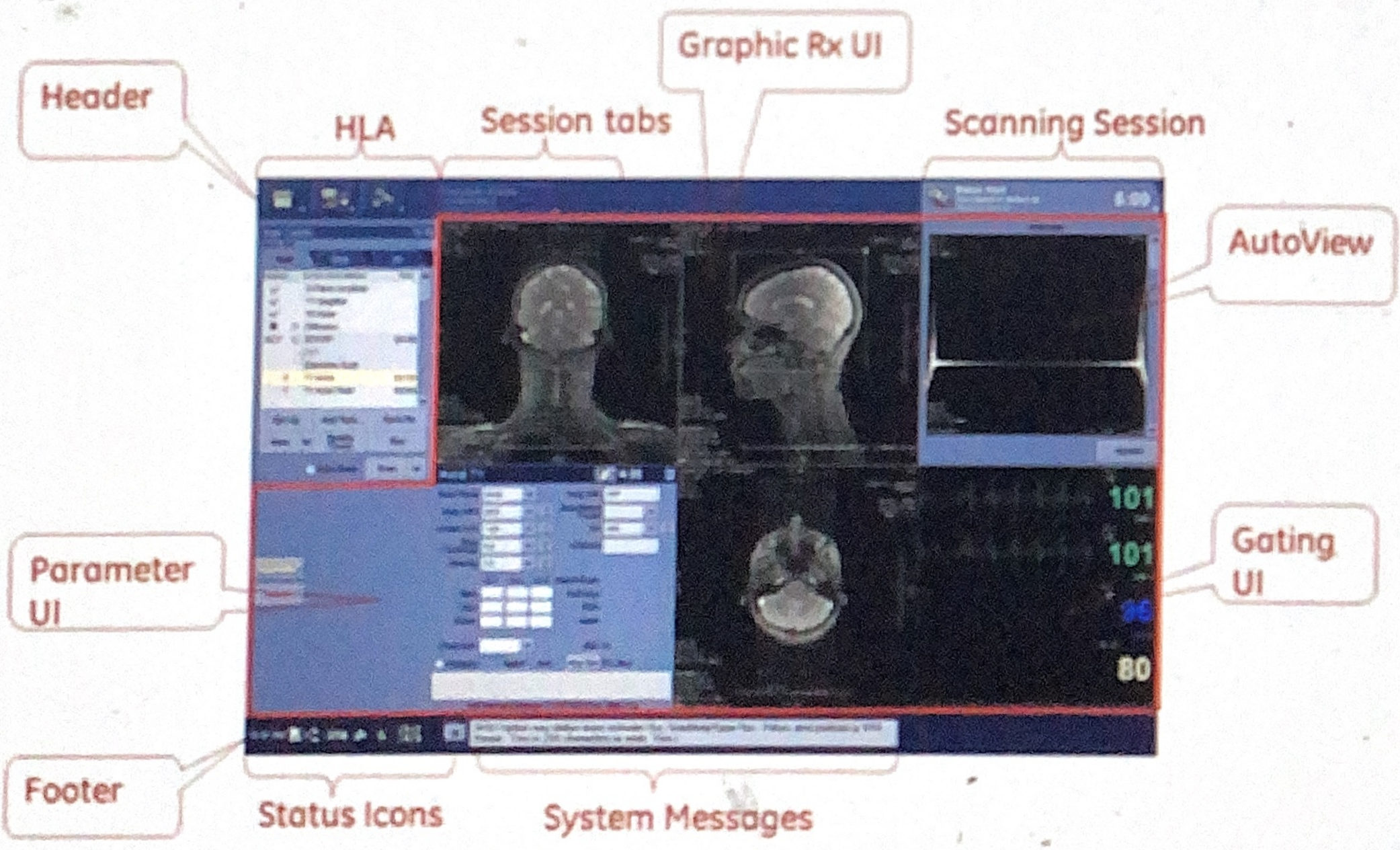
No specific hardware interface exists for this subsystem. All hardware interfaces are managed by the Linux OS for the host computer hardware interfaces (device drivers). The MRI hardware devices are controlled by other subsystems'.

2.5 User Interfaces

Figure 3 below shows the general layout of the screen where the user interface associated with the host software resides.

The host software user interface resides in three main sections, the gating display, the autoview display and the application area. Additional data, such as protocol notes can appear in the additional application space. The autoview and gating displays are always present, regardless of the desktop navigation choice.

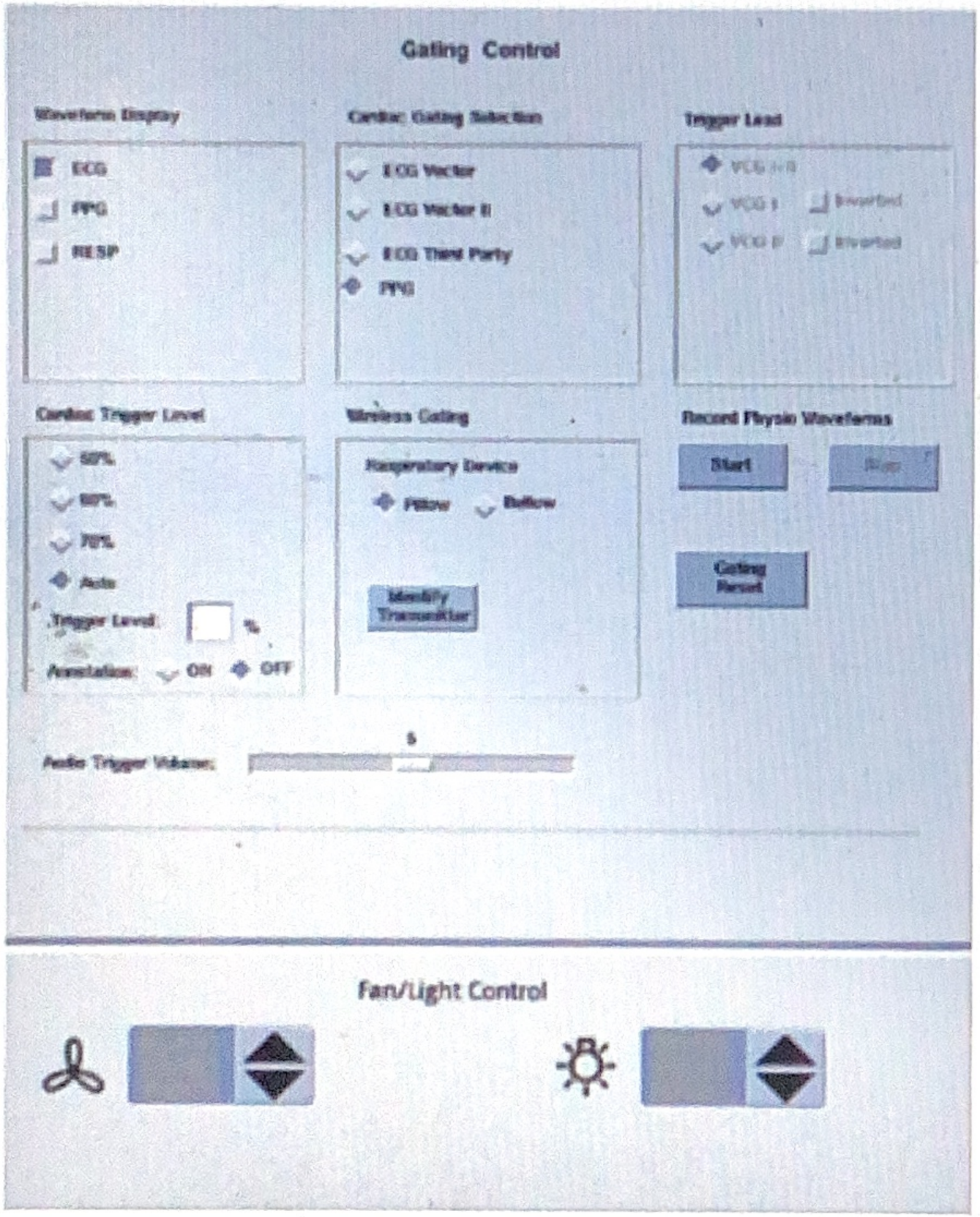
Figure 3 User Interface Screen Layout



2.5.1 Gating Control Interface

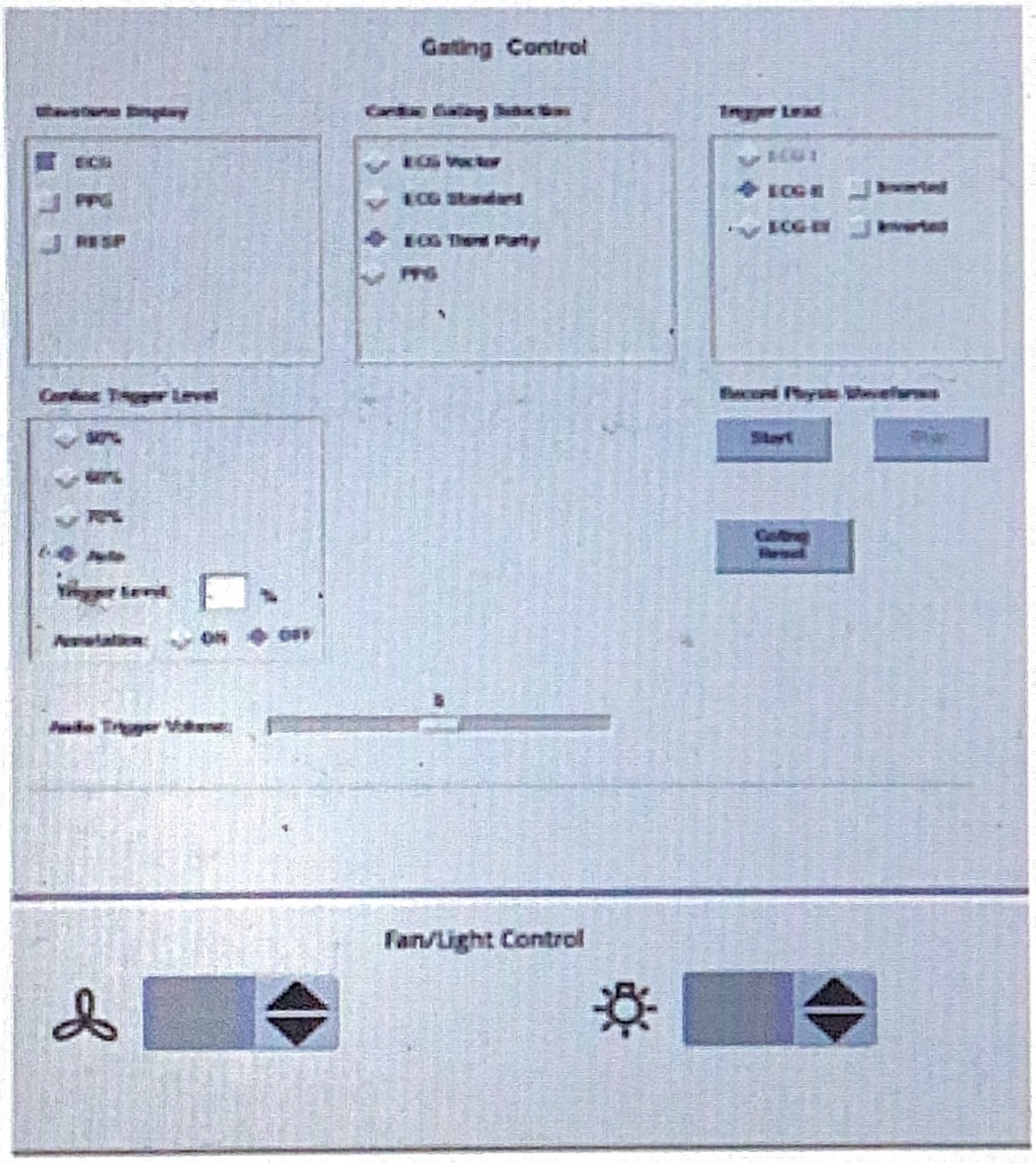
© 2022 by General Electric Company. All Rights Reserved.

2.5.1.1 Wireless Pac Type Gating Control



© 2022 by General Electric Company. All Rights Reserved.

2.5.1.2 Wired Pac Type Gating Control



2.6 Architectural Error Handling and Recovery

The error handling strategy for the MRI system consists of logging, automatic recovery, manual recovery and automatic system shutdown.

Each subsystem is responsible to detect and log errors as well as pass status back to the subsystem requesting the operation.

When feasible the subsystem should attempt recovery (i.e. retry, auto restart, etc.).

* Printed versions are For Reference Only *
Before using this document, consult MyWorkshop for the latest revision. GE Proprietary and Confidential Information.
Template Reference: 7.3.3-03MREG1, Design Description Template, DOC0600591

© 2022 by General Electric Company. All Rights Reserved.

When automatic recovery is not possible operator intervention is required. This interaction can range from forcing a partial system reset to retrying the operation. For example:

- Insufficient disk space to begin image acquisition, requiring the operator to remove unneeded images freeing sufficient space to proceed.
- During prescription the system will validate the operator input and provide suggested values to resolve incompatible parameter values.
- Table not at scan location, operator needs to advance the table then retry the operation.
- System Sequence and Control subsystem or recon subsystem fault requiring a subsystem reset. The operator is informed of the issue (the specific fault condition is logged) and the operator is requested to perform a system reset.

In rare cases a catastrophic system fault occurs, which causes an automatic shutdown of the system. The system will attempt to log the fault to facilitate subsequent repair and/or remediation.

2.7 Architectural Risk Control

Through the FMEA process, hazards (as identified in the MR RAC [32]) are associated with specific components within the host software architecture, see reference [31]. This information is captured in the *MR Software Structure and Safety Decomposition* spreadsheet, reference [33], and used to classify the software components. The software FMEAs have identified the following critical hazards to be associated with software:

1. RAC ID 2.200.30 - Excessive SAR (RF)
2. RAC ID 10.100.130 - Orientation or annotation of images (primarily relied upon)
3. RAC ID 10.100.142 – Sagittal image ordering (primarily relied upon)
4. RAC ID 10.100.170 - Data/Patient mismatch

Refer to the Host software FMEA for the serious and minor hazards associated with the software.

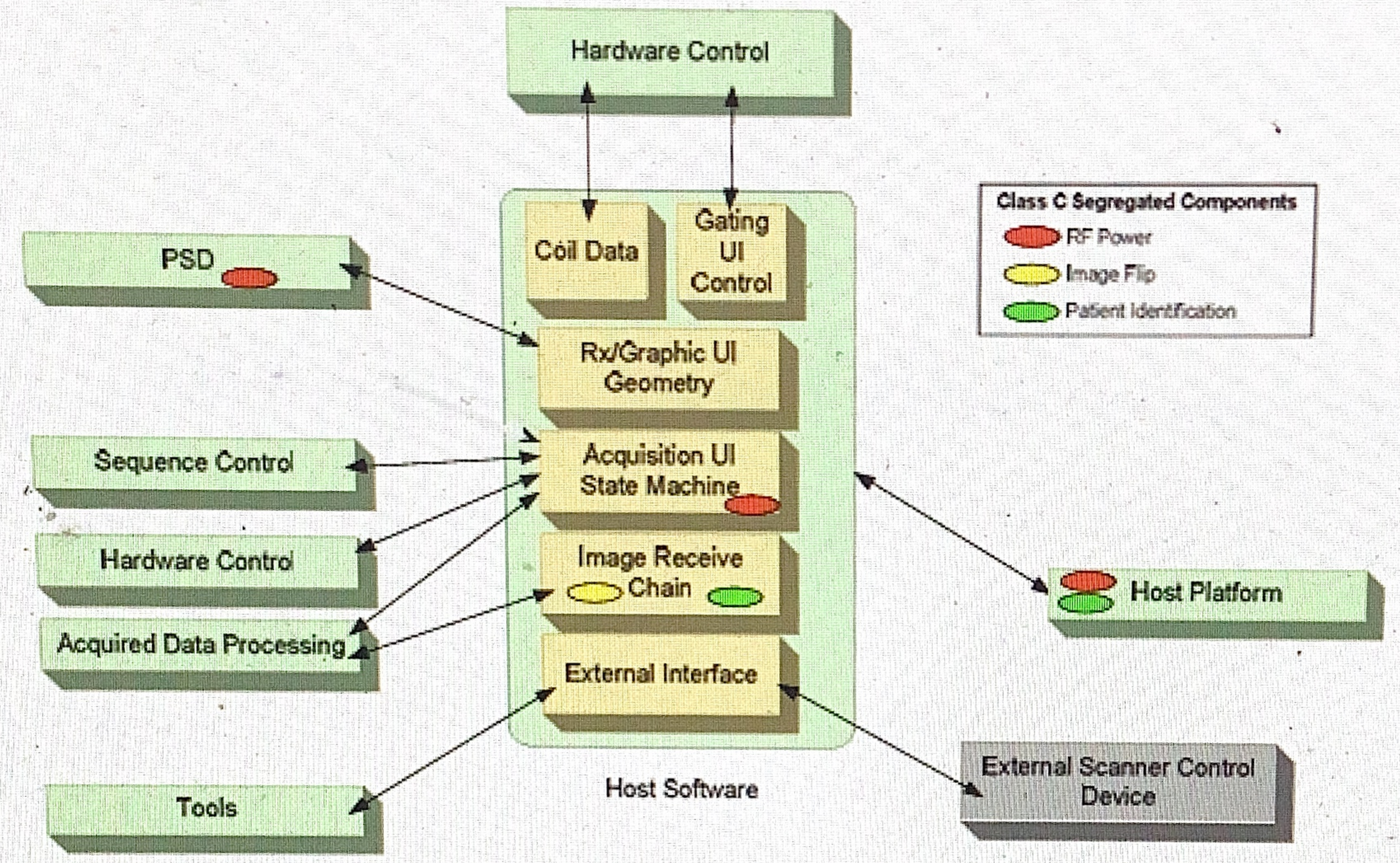
Due to the risk associated with the critical items, the MR Host software has been given the safety classification of Class-C. Although the current architecture was not initially designed to provide clear segregation or redundancy to address class C software risk control, this design has been modified to segregate the relevant software components related to the critical safety risks

Figure 4 class C components in Host Software identifies where the segregated Class-C components exist within the MR Host Software Sub-System

Figure 4 class C components in Host Software

* Printed versions are For Reference Only *
Before using this document, consult MyWorkshop for the latest revision.

© 2022 by General Electric Company. All Rights Reserved.



The software associated with the critical hazards has been segregated from the remainder of the software, resulting in a Class B or Class A designation of the other components. The division of the software from subsystems to units is contained in the segregation structure - [5]. All critical data passed between components will be encapsulated with a checksum so that the data recipient can detect data corruption and take the appropriate action to ensure system, operator and patient safety. In some cases components in the current architecture are legacy components and were developed before GEHC_GQP_10.04 Software Development Lifecycle Procedure was effective (June 5, 2009) and is thus considered legacy code according to the MR SDLC WI (DOC0593175). These software components are considered class C and no segregation has been performed.

Details of the segregation, for the above mentioned hazards, are contained in the sections below.

2.7.1 Software Segregation: RF Power Protection

The RF power management software has been segregated from the surrounding software by means of separation, encapsulation and acknowledgement.

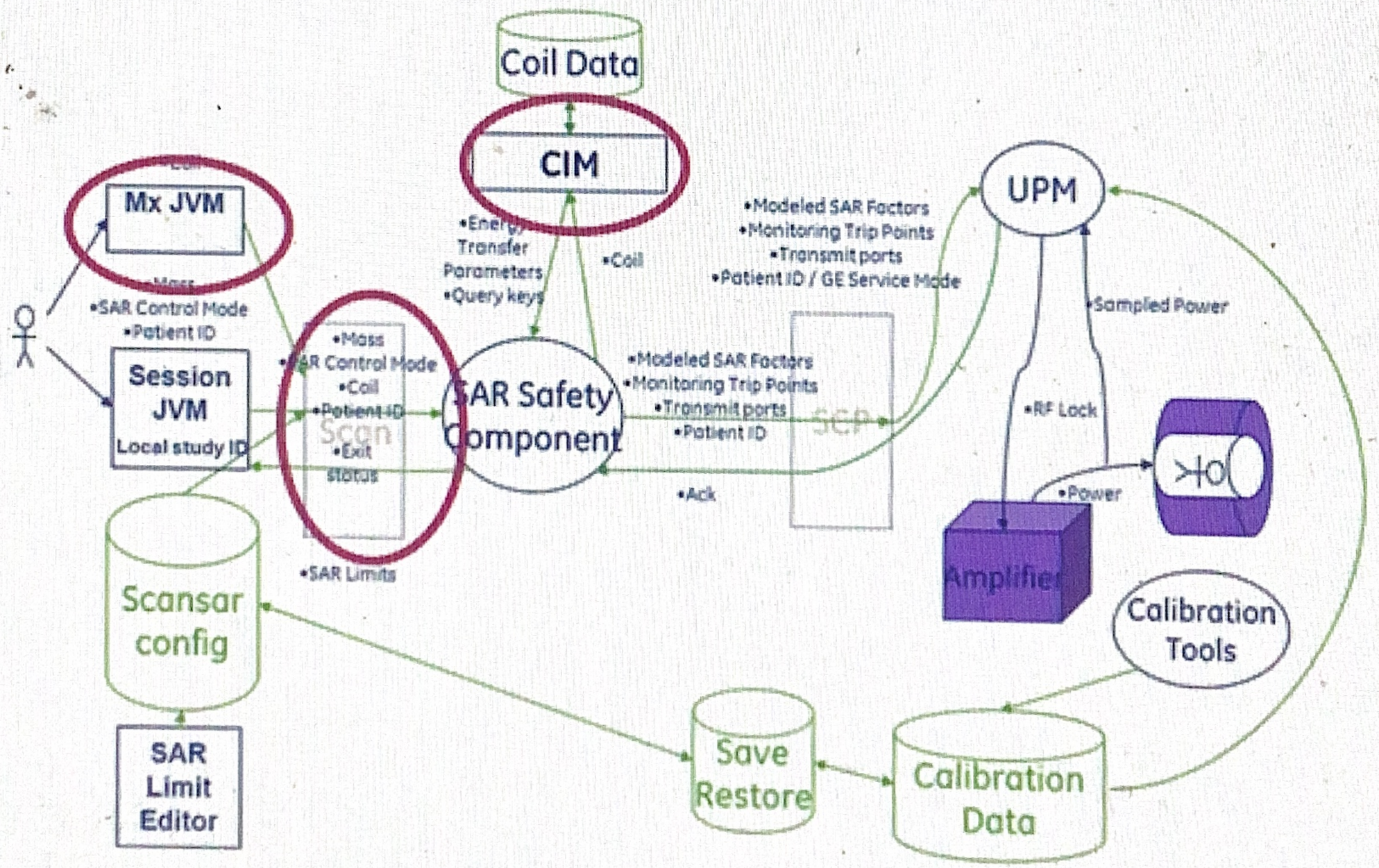
The power monitor is a separate component within the system responsible to monitor the output RF power and to disable the RF transmitter if the transmitted power exceeds the safety settings. The

* Printed versions are For Reference Only *
Before using this document, consult MyWorkshop for the latest revision. GE Proprietary and Confidential Information.
Template Reference: 7.3.3-03MRF01 Design Description Template DOC0600591

© 2022 by General Electric Company. All Rights Reserved.

receipt of data parameters transmitted to the power monitor is acknowledged to ensure the proper parameter set is being enforced. The critical RF power parameters are encapsulated via checksums to prevent inadvertent alteration. This data is checked at the endpoints using this information. The SAR Safety component, which was added to compute the setup parameters for the power monitor, runs as an independent component to prevent interaction with the other operations performed during scan prescription. This component validates its input data to ensure that the input has not been altered. Additionally, it verifies the consistency keys within the coil data to ensure the data belongs to the correct coil. Figure 5, below, depicts the dataflow within the system associated with this activity. Host Software owned components are marked red.

Figure 5 RF Power Setting Data Flow



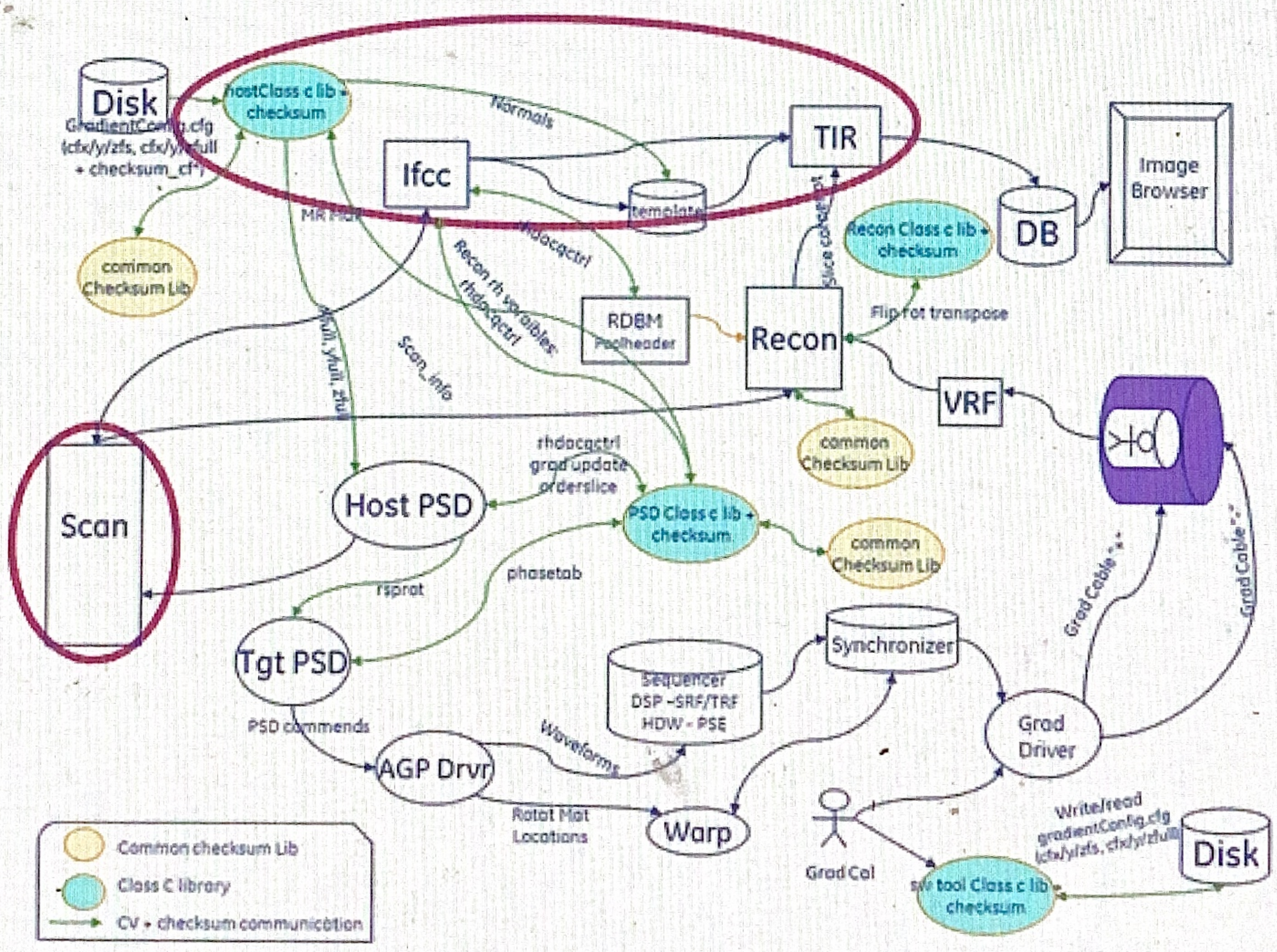
The software tools associated with RF calibration are also considered Class-C. When the tool completes the system calibration, the calibration data will be signed with a checksum to ensure that the data integrity is maintained.

© 2022 by General Electric Company. All Rights Reserved.

2.7.2 Software Segregation: Image Flip Protection

The hazard associated with unintended image flips (right/left, top/bottom) are managed via isolation, consistency checks, and checksum encapsulation of the data when it is being transferred between the various system components. In order to protect from unintended image flips the critical components that could cause image flip within each subsystem are isolated. Checksums are used to protect the data integrity as it passes thought the various components within the subsystems.
The critical computations have been isolated into separate libraries (PSD and Host) to segregate them from the remainder of the code set. Data parameters which reside in system configuration files are also protected by means of checksums. Since the image header data follows a different path through the system than the pixel data, the process which combines the header and pixel data into the final image also verifies that the pixel data matches the expected spatial data for the image.
Figure 6, below, shows the critical data flow within the system related to image flip. Host Software owned components are marked red.

Figure 6 Data flow related to image orientation



* Printed versions are For Reference Only *
Before using this document, consult MyWorkshop for the latest revision. GE Proprietary and Confidential Information.
Template Reference: 7.3.3-03MRF01_Design Description Template_DOC0600591

© 2022 by General Electric Company. All Rights Reserved.

2.7.3 Software Segregation: Patient Identification Protection

The hazard related to patient identification is associated with failure to properly associate the patient information (patient ID, name, data of birth) with the image data. A failure of this capability would result with patient data being associated with the wrong patient. The software components that could lead to incorrect association between the patient information and the image have been segregated from the software. The data passing between various subsystems will be protected through the use of checksums.

The subsystem which creates the patient information will generate a checksum of the data. This checksum will follow the data and be rechecked at image installation time to ensure data integrity. Additional integrity checks verifying the exam, series and image numbers will be performed at the point where the pixel data is associated with the header data.

The reconstruction subsystem will clear all working memory between studies to ensure image data does not inadvertently cross study boundaries. Additionally, all accesses to working memory within the reconstruction subsystem will be validated to ensure the correct memory segments are being accessed.

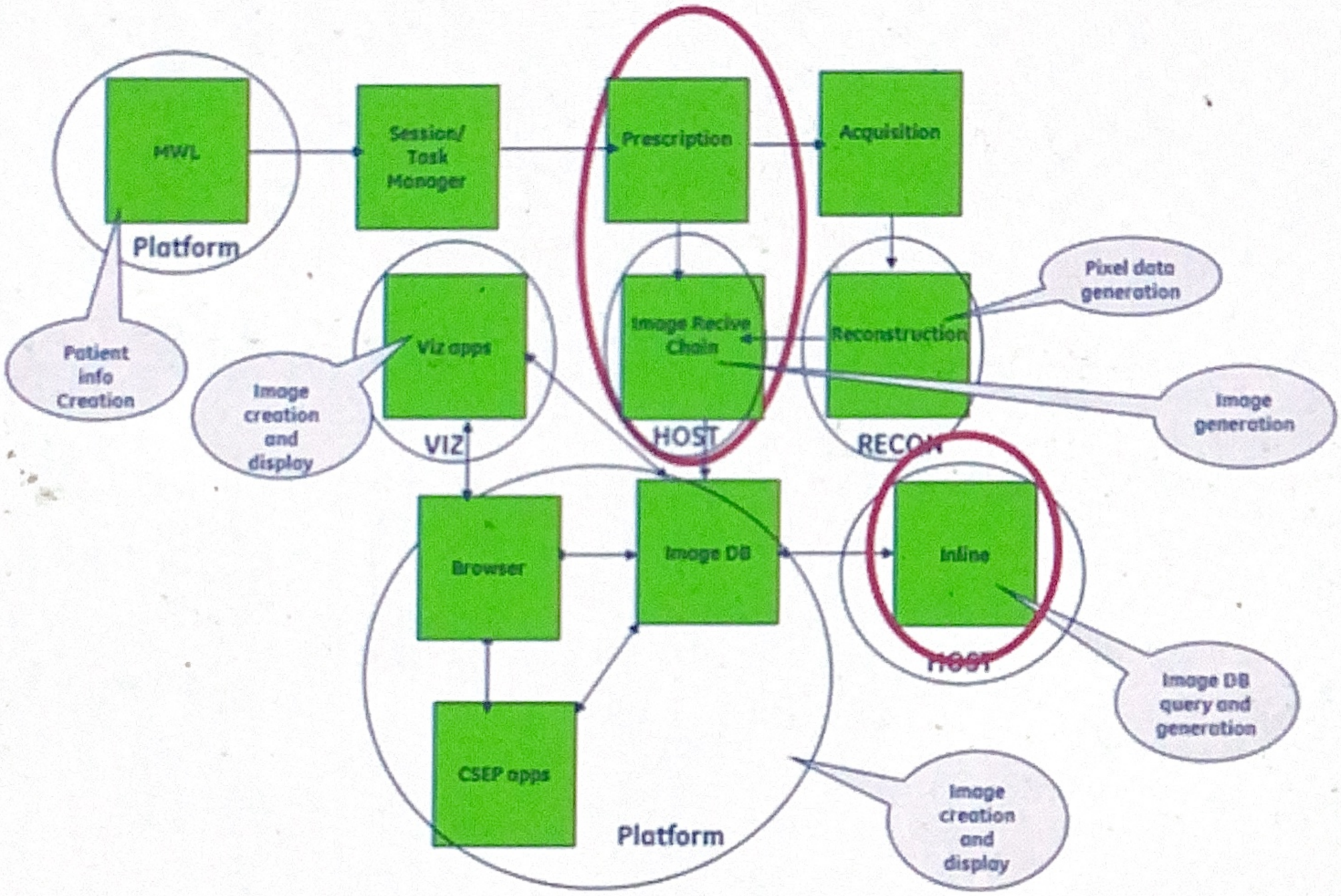
The data base subsystem will ensure that image checksums exist in the DICOM image preamble. Images received via media or network transfers will have checksums added prior to data base insertion. Images being installed from within the system will be checked against their respective checksum data.

The visualization and post processing software will use the embedded checksums to verify that the images have not been corrupted. Additionally, they will verify that the set of data received from a data base query match the expected patient.

Figure 7, below, show the data paths and interactions associated with the patient information tied to the image data. Host Software owned components are marked red.

Figure 7 Patient ID data paths

© 2022 by General Electric Company. All Rights Reserved.



2.8 *Existing Design Reusability*

The software architecture described within this document is common across all DV hardware based MRI systems. However, reusability in general will be software component driven, dependent on specific features (HW & SW) associated within a specific product implementation. Specifics on reusability will be covered in further detail within the component/unit SWDD documents.

Across both components and subsystems, one primary set of components designed for reusability provide the inter-process communication capabilities for the system. These communication channels are implemented using the following basic communications transports:

- TCP/IP
- Shared Memory
- Files
- Message Queues

* Printed versions are For Reference Only *
Before using this document, consult MyWorkshop for the latest revision. GE Proprietary and Confidential Information
Template Reference: 2.3.3-03MREF01, Design Description Template, DOC0600591

© 2022 by General Electric Company. All Rights Reserved.

These low level transports have been abstracted into the following high level communication APIs:

- MrMail/CM – custom packet based communication protocol supporting both Linux and VxWorks tasks supporting named lookup for endpoints. Built on-top of TCP/IP, RPC. MrMail is the high level API on the host; CM is the high level API on the VRE's and VxWorks boards.
- Corba – common object request broker
- Shared Memory – implemented via SVR4 shared memory or memory mapped files
- Message Queues – POSIX message queues
- CUP/IPC – inter-process communication protocol layered on top of TCP/IP, but with a simpler API.
- Memory Server – used to transport large blocks of data between processes residing on separate compute nodes (generally images but can be any large data block)
- SVAT Interface for automated scanner control
- MX framework communication interfaces: Mx event, Mx spider bus.
- GEOCOMM stands for **GE Object Communications** and is an object-based inter-process communication mechanism supporting both Java and C++ programming languages. It is Java centric – the developer writes Java code from which support Java methods and C++ code can be automatically generated. It also Supports sending & processing of objects

In general, each of the above communication APIs are used for point-to-point communications.

2.9 Future Considerations

In the future more hardware abstraction will be introduced within the software subsystems allowing a wider hardware base to be supported within the same software set. The current software subsystems primarily support multiple field strengths (magnet and frequency dependent components) on essentially similar hardware platforms.

3. Design Architecture

Chapter 3 provides a high-level overview of how the functionality and responsibilities of the MR Software System described by this SWDD were partitioned. The partitioning, except as noted, follows the hardware partitioning of the MRI system.

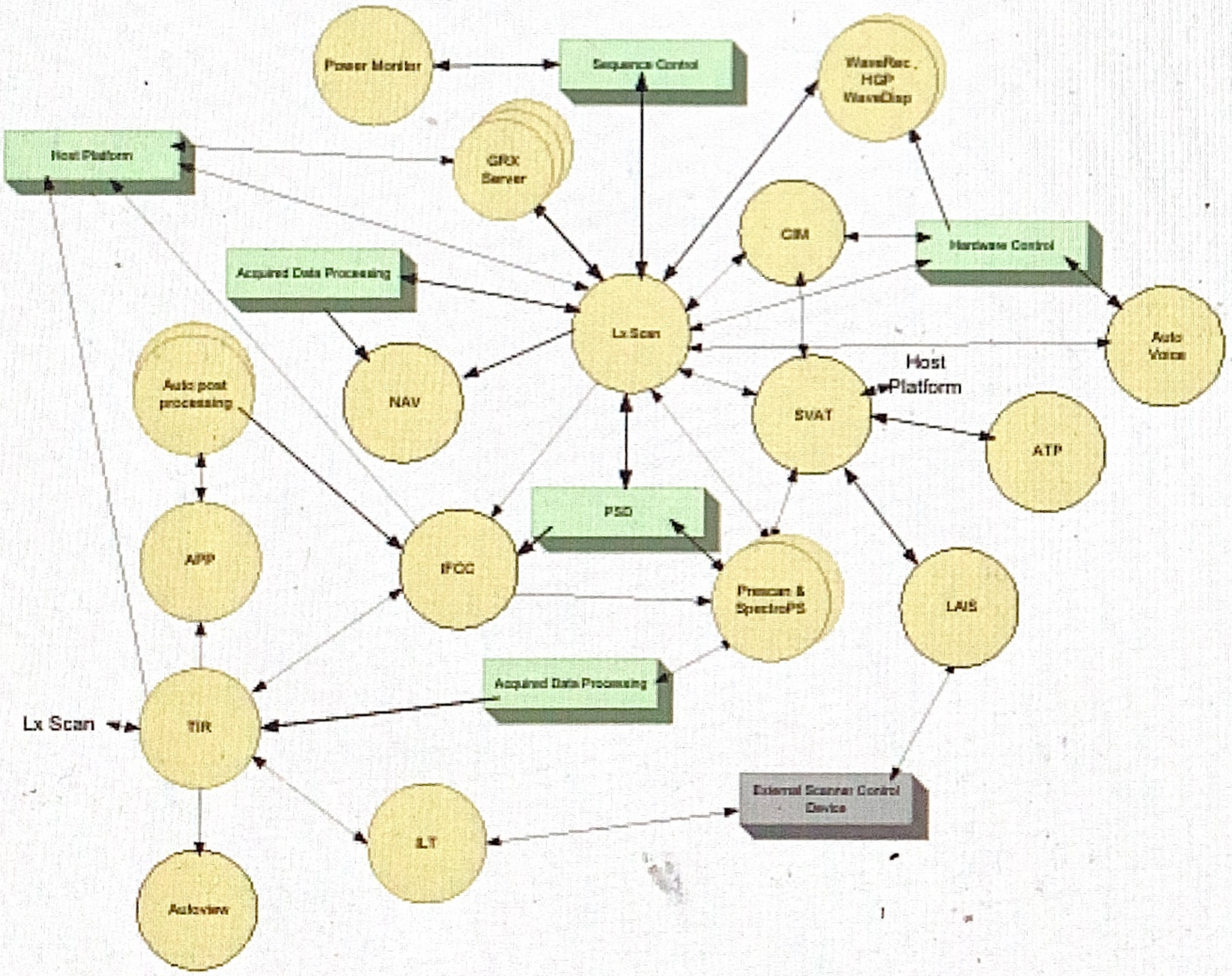
© 2022 by General Electric Company. All Rights Reserved.

3.1 *Internal Functional or Object design overview*

The MRI host subsystem software has been divided into 24 primary software components. Figure 8 through Figure 11 below depicts these components and shows their communication relationships for the four primary operational modes of the MRI system (Lx-Batch, Lx-RTIA, Lx-FMRI and Mx).

Figure 8 illustrates the software architecture for the Lx batch mode operation of the system. For batch mode series acquisition, the operator prescribes the imaging control parameters for series to be acquired, saves the prescription for later execution, and then executes the acquisition. While the acquisition is in progress the operator is free to prescribe additional series or perform post processing or image viewing operations. The system is capable of automatically acquiring (sequentially) a set of prescribed acquisition tasks when running with auto-scan enabled.

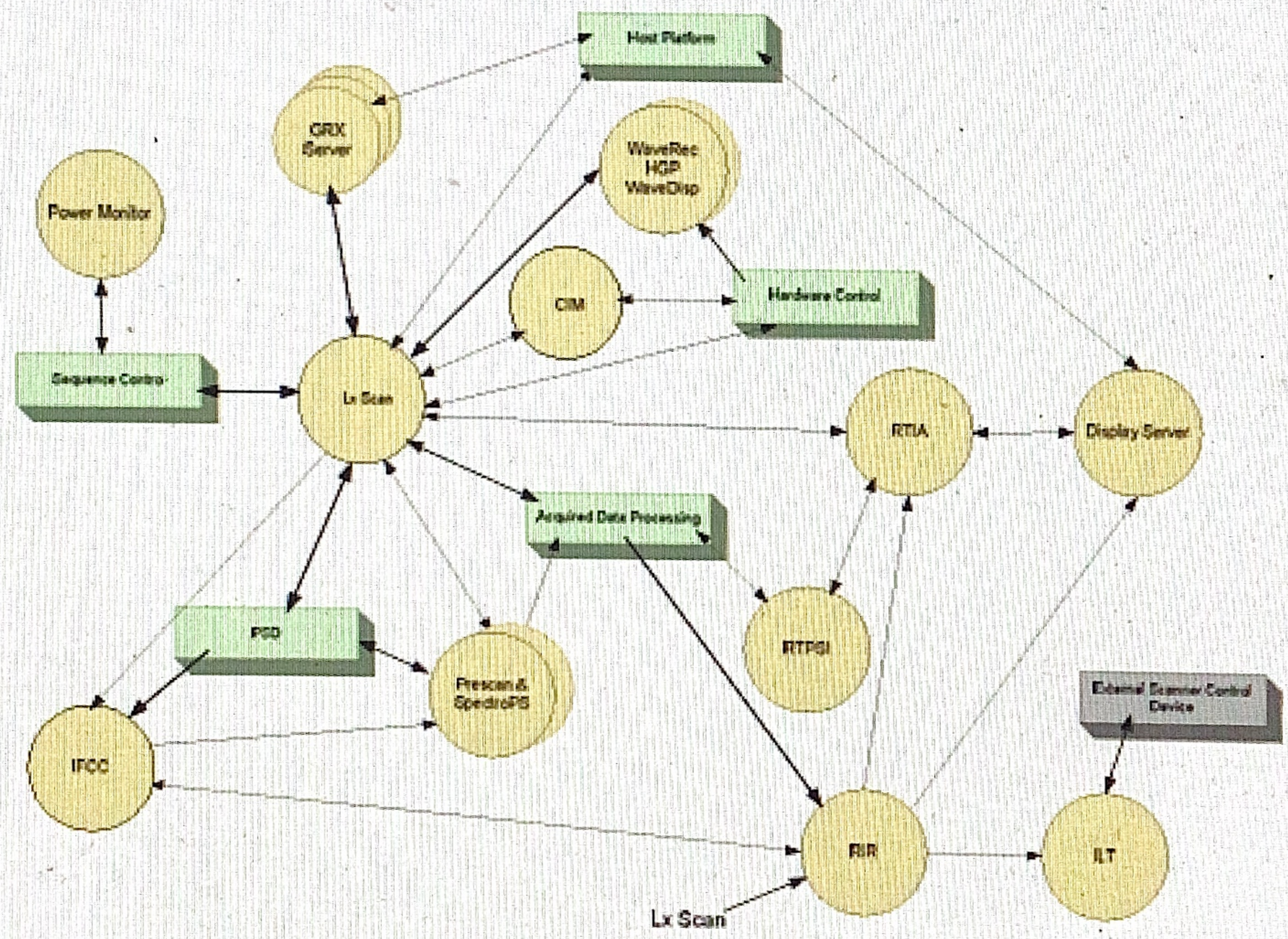
Figure 8 Host Software Architectural Breakdown – Lx Batch



© 2022 by General Electric Company. All Rights Reserved.

Figure 9 illustrates the software architecture for the Lx RTIA mode operation of the system. For real-time mode, the operator prescribes the imaging control parameters for images to be acquired, saves the prescription, and then executes the acquisition. This opens RTIA acquisition UI. This user interface allows users to modify scanning parameters on fly and view acquired images. It also enables real-time navigation. It provides mechanism to operate in review mode where it is possible to pause scanning, review acquired images and save selected images into database. It is not possible to prescribe or scan any other tasks until we end RTIA scanning and close UI.

Figure 9 Host Software Architectural Breakdown – Lx RTIA



* Printed versions are For Reference Only *
Before using this document, consult MyWorkshop for the latest revision. GE Proprietary and Confidential Information.
Template Reference: 7.3.3-03MREG1 Design Description Template, DOC0600591